

We can see that the dates and their high temperatures are stored in columns 2 and 4. To explore this data, we'll process each row of data in *sitka_weather_07-2021_simple.csv* and extract the values with the indexes 2 and 4.

Extracting and Reading Data

Now that we know which columns of data we need, let's read in some of that data. First, we'll read in the high temperature for each day:

```
sitka_highs.py  --snip--
reader = csv.reader(lines)
header_row = next(reader)

# Extract high temperatures.
❶ highs = []
❷ for row in reader:
❸     high = int(row[4])
    highs.append(high)

print(highs)
```

We make an empty list called *highs* ❶ and then loop through the remaining rows in the file ❷. The reader object continues from where it left off in the CSV file and automatically returns each line following its current position. Because we've already read the header row, the loop will begin at the second line where the actual data begins. On each pass through the loop we pull the data from index 4, corresponding to the header *TMAX*, and assign it to the variable *high* ❸. We use the *int()* function to convert the data, which is stored as a string, to a numerical format so we can use it. We then append this value to *highs*.

The following listing shows the data now stored in *highs*:

```
[61, 60, 66, 60, 65, 59, 58, 58, 57, 60, 60, 60, 57, 58, 60, 61, 63, 63, 70,
64, 59, 63, 61, 58, 59, 64, 62, 70, 70, 73, 66]
```

We've extracted the high temperature for each date and stored each value in a list. Now let's create a visualization of this data.

Plotting Data in a Temperature Chart

To visualize the temperature data we have, we'll first create a simple plot of the daily highs using Matplotlib, as shown here:

```
sitka_highs.py  from pathlib import Path
import csv

import matplotlib.pyplot as plt

path = Path('weather_data/sitka_weather_07-2021_simple.csv')
lines = path.read_text().splitlines()
--snip--
```